

1. Discuss what information can be drawn from input and output characteristics of a transistor with the help of the case study of CE configuration. Also explain how this information is useful in designing an amplifier.

OR

Derive the expression on voltage gain, input impedance and output impedance of common gate amplifier.

2. The circuit shown in Fig. 1 is an amplifier using a p-n-p and an n-p-n transistor in parallel. The two transistors have identical characteristics. Find the expression for the voltage gain and the input resistance of the amplifier, using the simplified hybrid model.

OR

3. Assume all MOSFETs shown in the Fig 2 are in saturation. Calculate the small signal voltage gain of the circuit. (Consider CLM and Neglect Body bias). [5]
Derive the voltage gain of differential amplifier with resistive load using small signal analysis.

OR

4. Describe working of an emitter coupled differential amplifier and explain how it is affected when supplied with a constant current source. [5]
If an input signal V_i is impressed between gate and ground in Fig. 3. Find the amplification $A_v = V_o/V_i$. Apply Miller's theorem to the 50 K resistor. The FET parameters are $\mu = 30$ and $r_d = 5$ K. Neglect capacitances.

OR

Discuss about the frequency response of CS-stage using Miller's approximation [5]

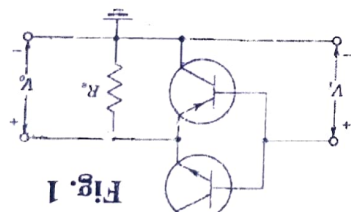


Fig. 1

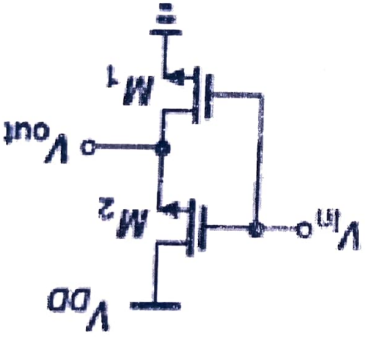


Fig. 2

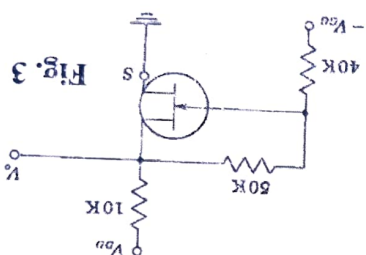


Fig. 3